

Biological Computationalism and Self Aware Networks

A Hybrid-Dynamics Mechanism Proposal for Substrate-Sensitive Cognition

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Abstract

Whether a physical process can be described computationally, whether reality is fundamentally computation, and whether consciousness depends on the physical realization of a computation are three different questions. This paper separates them and addresses the third. Biological computationalism argues that cognition and consciousness cannot be understood solely as abstract input-output functions because living systems compute through scale-inseparable, continuous and event-like processes enacted by electrochemical tissue. Self Aware Networks (SAN) is developed here as a bounded mechanism proposal for that substrate-sensitive view.

The paper begins from an ordinary fact: a neuron is already active before a new signal arrives. Its membrane voltage, dendritic conductances, inhibition, molecular state, recent history, metabolic condition, and relation to surrounding rhythms alter what the arriving signal can do. Biological computation is therefore modeled as a hybrid dynamical system in which continuous state evolves between discrete events and is changed by state-dependent jumps. A coupled-oscillator analysis then distinguishes random population addition from phase-compatible addition, defines the exact order parameter used by the scaling claim, and specifies when a noisy reset can contract phase dispersion.

Established evidence is kept separate from SAN interpretation. Dendritic computation, phase resetting, communication through coherence, KIBRA-PKM-zeta maintenance of late long-term potentiation and memory under studied conditions, CaMKII activity persistence despite molecular turnover, and ordinary electrochemical transduction provide empirical anchors. SAN adds the proposal that tonic activity supplies a receiver context, phasic departures carry receiver-relative differences, and recurrent receive-transform-project cycles may organize multiscale biological computation. The terms oscillatory canvas, Phase Wave Differential, and Neural Array Projection Oscillation Tomography are introduced only after those operations are explained in ordinary and inherited scientific language.

The result is not a proof that SAN explains consciousness, that only carbon-based systems can be conscious, or that neural synchrony is quantum coherence. It is a falsifiable research program specifying implementation variables, mathematical assumptions, measurement targets, failure conditions, and criteria an artificial realization would need to reproduce or replace.

Keywords: biological computation; hybrid dynamics; dendritic computation; neural oscillations; phase resetting; molecular memory; substrate dependence; Self Aware Networks; NAPOT; PWD

Scope and evidence boundary

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1. The Problem Before the Vocabulary

The same abstract algorithm can be written on paper, executed by silicon, embodied in a mechanical device, or approximated by a living circuit. Those implementations can agree on selected inputs and outputs while differing in latency, energy use, noise, adaptation, internal state, failure modes, and the transformations

available between input and output. For many engineering purposes those differences can be ignored. For consciousness, ignoring them is a hypothesis, not a settled result.

Three propositions must therefore be separated.

1. **Computational modelability:** a physical or neural process can be represented, predicted, or simulated by computational rules.
2. **Computational ontology:** what most fundamentally exists is computation.
3. **Consciousness-realizing substrate dependence:** the physical dynamics implementing a computation may be part of what makes the process conscious.

The first proposition does not entail the second. Neither settles the third. Saying that neural dynamics are computable is compatible with several ontologies, and saying that an artificial system is computationally sophisticated does not yet show that it preserves the causal organization relevant to biological consciousness.

The scientific problem is consequently more exact than a contest between “biology” and “software.” We need to identify which physical variables are merely replaceable details and which variables perform indispensable causal work. A defensible substrate-sensitive theory must specify those variables, show how they interact, state what an artificial equivalent would have to preserve, and expose the proposal to measurements that can fail.

2. The Established Biological-Computationalist Challenge

Milinkovic and Aru argue that current debates often assume computational functionalism: the view that the right abstract information-processing organization is sufficient regardless of physical substrate. Their biological-computationalist alternative emphasizes two features of living computation: scale inseparability and hybrid continuous-discrete dynamics [1]. Block independently frames the related question in terms of whether electrochemical realization contributes something that abstract computation leaves out [2].

This paper agrees with the research constraint without treating it as a completed consciousness theory. Living neural computation is enacted through:

- ionic concentration gradients and membrane currents;
- morphology-dependent dendritic integration;
- probabilistic vesicle release and receptor dynamics;
- inhibition, recurrent excitation, and neuromodulation;
- oscillatory timing, phase resetting, and traveling activity;
- molecular turnover, plasticity, repair, and homeostasis;
- metabolic supply and embodied sensory-action feedback.

The slogan preserved from the source manuscript - “the algorithm is the substrate” - is useful only if read carefully. It does not mean that every molecule is individually indispensable or that no nonbiological implementation could work. It means that the relevant algorithm cannot be specified solely by terminal input-output behavior while omitting the causal state transitions that produce that behavior.

3. A Plain-Language Operation

Consider a receiving neuron before an input arrives. The cell is not blank. Its dendritic branches have different conductances and synaptic histories. Inhibition may make one route less available than another. Membrane voltage changes the effect of an excitatory input. Recent spikes alter ion-channel availability.

Nearby populations may be rhythmically coordinated or dispersed. Molecular processes affect how long a resulting change persists.

An arriving event is therefore transformed in context:

existing receiver state

- + arriving event and its timing
- + local anatomy, inhibition, and metabolic condition
- > state-dependent transformation
- > changed output timing or probability
- > changed downstream receivers
- > recurrent sensory, bodily, or network return

This operation contains both continuous and event-like components. Membrane and molecular states evolve continuously over useful approximations of time. Spikes, release events, threshold crossings, and externally imposed perturbations create identifiable events. The event does not float above the biology; it changes a state whose physical composition and history matter.

That is the operation this paper calls **biological computation**: the selection, stabilization, and propagation of state changes through event-like transitions embedded in continuous, history-dependent biophysical dynamics.

4. Formal Hybrid-Dynamics Specification

Let $x(t)$ collect the continuous variables needed at the scale of a chosen experiment: membrane potentials, conductances, calcium concentrations, adaptation variables, molecular activities, and population phases. Let $u(t)$ be controlled or observed input, and let $\lambda(t)$ denote slowly changing context such as neuromodulation, metabolic state, or plasticity parameters. Between identified events t_k , the system evolves as

$$\dot{x}(t) = f(x(t), u(t), \lambda(t)), \quad t \neq t_k. \quad (1)$$

At an event e_k , the state is updated by a state-dependent jump map:

$$x(t_k^+) = J(x(t_k^-), e_k, \lambda(t_k)). \quad (2)$$

Measured output is not assumed to reveal the full state:

$$y(t) = h(x(t), \{e_k : t_k \leq t\}) + \epsilon(t), \quad (3)$$

where $\epsilon(t)$ collects measurement error and unmodeled variability. Equations 1-3 are not uniquely SAN equations; they are a standard hybrid-systems form used here to make the biological-computationalist claim operational. SAN enters through proposed choices of state variables, receiver-relative differences, recurrent architecture, and cross-scale observables.

This specification prevents two common simplifications. First, a spike cannot be treated as a context-free binary token because J depends on pre-event state. Second, a continuously varying field or molecular process cannot be assumed to determine cognition without identifying an event, transformation, or output through which it changes later behavior.

5. What SAN Adds, After the Operation Is Understood

SAN proposes that ongoing activity supplies a maintained physical context and that new content is expressed through departures from, and transformations of, that context. In the mature SAN vocabulary:

- the **oscillatory canvas** is the maintained tonic context created by ongoing cellular and population activity;
- a **Phase Wave Differential (PWD)** is a candidate receiver-relative change in timing, phase, amplitude, or state relation that alters what a receiving unit does next;
- **Neural Array Projection Oscillation Tomography (NAPOT)** is the proposed recurrent receive-transform-project architecture by which partial state estimates may be exchanged and updated across neural arrays.

These terms are not synonyms for phase coding, traveling waves, predictive coding, or communication through coherence. Those inherited concepts describe important components. The proposed delta is the joined causal sequence:

an already-active receiver

- > detects a state- and route-relative difference
- > transforms that difference according to local structure and history
- > projects an updated event pattern
- > changes other receivers and the distributed model
- > receives recurrent bodily, sensory, and network consequences

PWD is thus not defined merely as “a phase code.” The candidate variable is relative to a receiver, a path delay, an expected relation, and a current context. A minimal phase component can be written

$$d_{ij,k} = \text{wrap} \left[\theta_j(t_k - \tau_{ij}) - \theta_i(t_k) - \phi_{ij}^{\text{exp}} \right], \quad (4)$$

where j is the sending unit or population, i is the receiver, τ_{ij} is path delay, and ϕ_{ij}^{exp} is an experimentally estimated expected phase relation. Equation 4 becomes informative only when changes in $d_{ij,k}$ predict or causally affect a receiver update beyond established alternatives. Without direction, delay, baseline, and intervention, it remains a candidate descriptor.

NAPOT similarly becomes scientific only if a recurrent neural reconstruction can be identified, perturbed, and distinguished from ordinary decoding or correlation. Successful stimulus decoding alone does not establish tomography, a rendered observer, or consciousness.

6. The Population-Scaling Claim, Corrected and Made Exact

The earlier manuscript used the useful shorthand $a\sqrt{N}$ for incoherent addition and aN for aligned addition. The exact bridge requires the assumptions to be visible.

For N equal-amplitude contributions with phases $\theta_j(t)$, define the complex order parameter

$$Z(t) = \frac{1}{N} \sum_{j=1}^N e^{i\theta_j(t)} = r(t)e^{i\psi(t)}, \quad 0 \leq r(t) \leq 1. \quad (5)$$

The magnitude of their vector sum is then exactly

$$A_{\text{pop}}(t) = a \left| \sum_{j=1}^N e^{i\theta_j(t)} \right| = aNr(t). \quad (6)$$

If the phases are independent and uniformly distributed, the expected squared magnitude is

$$\begin{aligned} \mathbb{E}[A_{\text{pop}}^2] &= a^2 N, \\ A_{\text{RMS}} &= a\sqrt{N}. \end{aligned} \quad (7)$$

If all phases align, $r = 1$ and $A_{\text{pop}} = aN$. These are endpoints of a phasor model, not a universal law for every electrophysiological measurement. Unequal amplitudes, spatial cancellation, volume conduction, filtering, geometry, correlations, and the measurement operator must be included in an experiment-specific forward model.

A simple reset relative to the current mean phase can be modeled by

$$\begin{aligned} \delta_i^+ &= (1 - \alpha)\delta_i^- + \eta_i, \\ \delta_i &= \text{wrap}(\theta_i - \psi), \\ 0 &\leq \alpha \leq 1, \end{aligned} \quad (8)$$

where η_i is reset noise. If η_i is independent, zero mean, and has variance σ_η^2 , then

$$\text{Var}(\delta^+) = (1 - \alpha)^2 \text{Var}(\delta^-) + \sigma_\eta^2. \quad (9)$$

The reset contracts phase variance only when

$$\sigma_\eta^2 < \alpha(2 - \alpha) \text{Var}(\delta^-). \quad (10)$$

Equations 5-10 turn “alignment amplifies a signal” into a bounded model. They do not show that coherence is always beneficial. Strong global coherence can reduce representational differentiation, and pathological hypersynchrony is associated with some seizure dynamics. Useful computation may require structured coexistence of local coordination, phase dispersion, interference, and transitions among them. SAN therefore treats coherence and splay or decoherent relations as complementary organizational states rather than success and failure labels.

7. Thermal Noise, Transduction, and Amplification

Warm biological tissue is noisy. Tegmark’s analysis argues that particular long-lived, macroscopic quantum superpositions proposed for neural computation would decohere far faster than ordinary neural timescales [8]. This paper does not require those states. It also does not turn that result into the broader claim that quantum mechanics is irrelevant to chemistry or biology.

The experimentally useful chain is classical at the neural and circuit scales unless a separate bridge is demonstrated:

local molecular or synaptic perturbation

- > receptor, channel, vesicle, or membrane transduction
- > state-dependent amplification or suppression
- > changed spike timing or release probability
- > recurrent and population-level coordination
- > plastic or behavioral consequence

Every arrow can fail. A seed may be too small or nonspecific. An amplifier may raise both signal and noise. Phase alignment may be epiphenomenal. A persistent molecular change may not alter network timing. A measurable network change may not affect task performance. The thermal-noise response is therefore not “noise does not matter.” It is a demand for a quantitative signal path, controls, and an intervention showing that the path contributes to the claimed computation.

8. Empirical Anchors and Their Boundaries

8.1 Dendrites and state-dependent transformation

Dendrites implement linear and nonlinear transformations whose effects depend on morphology, conductance, synaptic placement, and state [3]. This supports treating the neuron as more than a context-free point unit. It does not establish SAN’s full receiver / projector analogy or NAPOT.

8.2 Oscillatory routing and phase reset

Communication-through-coherence proposes that phase relations can influence effective communication between populations [4]. Phase resetting has been studied as a candidate mechanism for coordinating large-scale networks during attention and goal-directed behavior [5]. These literatures support measurable timing and routing variables. They do not by themselves prove that synchrony is consciousness or that a particular PWD exists.

8.3 Molecular persistence across turnover

Tsokas and colleagues report that KIBRA-PKM-zeta complexes maintain late-LTP and long-term spatial memory under their studied conditions, including one-month memory despite PKM-zeta turnover; targeted antagonists disrupted established late-LTP and memory while relevant controls constrained interpretation [6]. Lee, Chen, and Nicoll report stable CaMKII activity in hippocampal synapses over two weeks in slice culture despite protein turnover and propose activity transfer to newly synthesized CaMKII [7].

These results are important because they show how selected biological state can persist even though component molecules turn over. They do not establish that either pathway is the sole memory substrate, that all memories use the same mechanism, or that molecular persistence necessarily controls oscillatory timing. The last link remains an experimental question.

8.4 Robustness and homeostatic degeneracy

Biological circuits can preserve function despite variability in individual components through compensation and homeostatic regulation [11]. This complicates simple lesion logic: one pathway may be causal under ordinary conditions yet replaceable after chronic adaptation. Acute, chronic, rescue, and matched-control interventions should therefore be distinguished.

9. A Multiscale Mechanism Map

Scale	Established operation	SAN proposal and discriminating question
Molecular	Kinase, scaffold, receptor, channel, and calcium dynamics alter persistence and excitability.	Does a selected molecular state predict or causally change a later receiver-relative timing update?
Dendritic and cellular	Morphology and voltage-dependent conductances transform spatially and temporally distributed input.	Does the receiver’s prior state determine the proposed differential and its downstream projection?
Synaptic	Release probability, receptor state, and plasticity update local transmission.	Can a bounded input-transform-output chain distinguish PWD from ordinary gain or phase coding?
Circuit	Inhibition, recurrence, and phase reset alter routing and population state.	Does intervention contract or expand phase dispersion and change task-relevant routing as predicted?
Network	Oscillations and traveling activity coordinate distributed populations.	Is there an identifiable recurrent reconstruction rather than stimulus decoding alone?
Embodied loop	Movement changes sensory evidence and bodily state.	Does closed-loop perturbation alter the distributed estimate and subsequent action in the predicted direction?

The proposed NAPOT program is the conjunction of these levels, not a claim that one frequency or one molecule explains consciousness.

10. Experimental Programs and Failure Conditions

10.1 Phase-reset intervention

Estimate θ_i , r , phase dispersion, directionality, and task-relevant output before and after a temporally precise inhibitory or sensory reset. Compare Equation 9 with observed dispersion. A successful fit without behavioral or routing consequences supports oscillator dynamics but not biological computation. The stronger claim requires a preregistered downstream effect and a timing-specific intervention.

10.2 Molecular-to-network persistence

Perturb a molecular persistence mechanism acutely, chronically, and with matched rescue. Measure synaptic efficacy, molecular state, phase statistics, recurrence, and behavior. The SAN bridge weakens if molecular effects occur without the predicted network-state consequence. It also weakens if the network effect is fully explained by a simpler excitability or performance confound.

10.3 Receiver-relative difference

Estimate path delay and expected phase relation from training data, freeze the estimator, and test Equation 4 on held-out trials. Compare against firing rate, univariate phase, amplitude, conventional phase-amplitude coupling, predictive-coding error, and traveling-wave baselines. A PWD claim requires incremental prediction or intervention evidence, not a renamed correlation.

10.4 Recurrent reconstruction

Use a closed-loop sensory-motor task in which partial evidence, viewpoint, or body state changes over time. Test whether distributed neural activity reconstructs latent relations across perturbations, whether the reconstruction is receiver-relative, and whether targeted disruption produces the predicted perceptual or action error. Decoding object identity is insufficient; the key test is an intervention on the proposed receive-transform-project cycle.

10.5 Artificial realization

Construct two systems with matched terminal behavior but different internal causal organization. One preserves the identified delays, continuous state, receiver-dependent jumps, plasticity, noise response, energy constraints, and closed-loop embodiment; the other abstracts them away. If both reproduce every discriminating result, the claimed substrate requirement narrows. If only the causally faithful realization does, the result supports implementation sensitivity without proving that biology is uniquely conscious.

11. Relation to Artificial Consciousness

This paper does not infer that an artificial system is conscious because it imitates human language, nor does it infer that artificial consciousness is impossible because present computers are not biological. It changes the burden of equivalence.

An artificial candidate should specify which biological operations it reproduces, which it replaces, and why the replacement is causally equivalent. Terminal behavioral similarity alone is weaker than:

- state correspondence across intervention;
- preservation of relevant transition structure;
- matched delay and recurrence;
- matched noise and perturbation response;
- matched learning and persistence behavior;
- matched embodied feedback where the claim requires it.

This is compatible with multiple realizability, but not with unrestricted multiple realizability. A different substrate may succeed if it instantiates a demonstrated functional, dynamical, or causal equivalent of the relevant operation.

12. Claim and Evidence Ledger

Claim ID and disposition	Supported claim	Boundary or failure condition
BC-COMP-01 - logical distinction	Computational modelability does not entail computational ontology.	Does not decide whether computational ontology is true.
BC-SUB-01 - research constraint	Abstract input-output equivalence may omit consciousness-relevant realization dynamics.	Requires identified variables and intervention tests.
BC-HYB-01 - formal model	Biological computation can be represented as continuous flow plus state-dependent events.	The chosen state and jump maps must be fitted and compared with alternatives.
BC-DEND-01 - established background	Dendrites perform state- and morphology-dependent transformations.	Does not establish NAPOT.
BC-OSC-01 - established framework	Phase relations and resets can influence effective network coordination.	Does not make all synchrony functional or conscious.

Claim ID and disposition	Supported claim	Boundary or failure condition
BC-MEM-01 - bounded empirical result	KIBRA-PKM-zeta interaction maintains late-LTP and memory under studied conditions.	Not a universal or exclusive memory mechanism.
BC-MEM-02 - bounded empirical result	CaMKII activity can remain stable across molecular turnover in the cited preparation.	Generalization beyond the preparation requires evidence.
BC-SCALE-01 - mathematical identity	Equal-amplitude phasors sum to aNr ; random phases have RMS scale $a\sqrt{N}$.	Forward measurement geometry and unequal amplitudes must be modeled.
BC-SAN-01 - mechanism proposal	Tonic context and phasic receiver-relative changes may organize biological computation.	Requires incremental predictive or causal evidence.
BC-PWD-01 - candidate discriminator	Direction-, delay-, baseline-, and receiver-conditioned differences may predict state updates.	Fails if conventional variables explain the result.
BC-NAPOT-01 - architecture proposal	Recurrent arrays may receive, transform, project, and update distributed reconstructions.	Decoding alone is insufficient; perturbational reconstruction evidence is required.
BC-AI-01 - scope condition	Artificial realization remains possible if relevant causal organization is reproduced or replaced by an equivalent.	Neither carbon exclusivity nor substrate indifference is assumed.

13. SAN Genealogy Relevant to This Paper

The present synthesis did not appear all at once, and later terminology must not be backdated into earlier sources.

Date	Source stage	Bounded contribution to the present paper
2011	Publicly described predictive-self and whole-neocortex records, with historical custody boundaries documented separately.	Functional ancestors of distributed self-modeling and no hidden observer; not NAPOT or PWD.
2012	EEG-mediated recursive light / sound feedback, coordinated neural-wave language, and the mind-projector / world-prediction account.	Oscillatory feedback and internally constructed world-model ancestors; mature phase-differential mathematics remains later.
2017	Neural Lace Podcast and related public writing on spatial computation, a changing 3D/4D ionic or electromagnetic matrix, dendrites as local computers, coincidence as an information bit, and alternating feedback / broadcast communication.	Public hybrid, spatial, dendritic, energetic, and receiver-transform genealogy.
2022	Immutable SAN Git history, GitHub Wiki, whitepaper development, and public NAPOT explainers.	Mature receive-transform-project arrays, distributed observer, PWD, NAPOT, COT, tonic / phasic organization, and multiscale rendering proposals.
2024	<i>Bridging Molecular Mechanisms and Neural Oscillatory Dynamics.</i>	Expanded molecular, plasticity, oscillatory, and scale-bridge synthesis.
2025	SAN-SIT bridge work and TCNR development.	Later extensions of multiscale computation; not historical proof of the earlier mechanism.

Date	Source stage	Bounded contribution to the present paper
2026	Response to Milinkovic and Aru, source recovery, formal hybrid specification, and this Version 2 synthesis.	Current explicit biological-computationalist framing and falsification program.

The authoritative atom-level dates, source grades, hashes, and public / private boundaries remain in the SAN genealogy and source ledgers. This table is a paper-local navigation layer, not a substitute for them.

14. Source, Attribution, and Intent-Recovery Boundaries

Draft 6 is preserved as the immutable predecessor for this revision. Version 2 retains its scope exclusions, operational definition, thermal-noise treatment, scale bridge, molecular anchors, mechanism map, falsifiers, claim ledger, cross-paper dependencies, source ledger, references, promotion scan, and conclusion. It strengthens rather than silently replaces those components.

The source classes are:

- **primary public literature:** scientific claims attributable to published sources;
- **public Micah genealogy:** dated public recordings, articles, repository history, and videos;
- **private authored source:** wording or reasoning used to recover authorial intent but not itself presented as public priority evidence;
- **model or assistant process evidence:** routing and criticism that must not be cited as scientific authority;
- **current synthesis:** language first fixed in this revision and never backdated.

Apparent errors in an earlier draft are treated as source-recovery questions before correction. The earlier wording remains recoverable; the revision records whether the intended claim was a typo, compressed analogy, incomplete derivation, older vocabulary, or genuinely unsupported proposition. Scientific correction does not authorize deletion of the historical narrative.

15. Cross-Paper Dependencies

Related work	Dependency retained in Version 2	Boundary
Paper 03, Oscillatory State Persistence and SVI	Oscillation and persistence vocabulary.	This paper keeps biological mechanisms primary.
Paper 09, Neuroscience Ontology and Symbolic SAN Graph	Claim labels, source classes, and graph-validation expectations.	An ontology pass is not biological validation.
Paper 11, Level 0 SIT Physics Letter	A separate physics mechanism and evidence boundary.	No gravity-equals-biology or neural-quantum-coherence inference enters this paper.
Paper 13, Experimental Collaboration Pitch	Downstream apparatus and protocol development.	Its incompleteness does not prevent this paper from stating bounded experiments.
Paper 14, Cross-Disciplinary SIT Translation Guide	Proof, model, analogy, conjecture, and prohibited-promotion language.	Translation is not derivation.
Paper 16, Provenance and Source-Chain Auditability	Public / private source handling and date-type separation.	Provenance is not empirical validation.

16. Limitations

The framework remains underdetermined in several ways. There is no unique state vector for biological computation across scales. The order parameter may obscure meaningful subpopulation structure. Molecular persistence may support memory without controlling the proposed oscillatory variables. Directionality estimates can be confounded by common input, volume conduction, and model choice. Receiver-relative phase differences may add no predictive information beyond established measures. Recurrent reconstruction may be explainable without tomography. Consciousness may depend on variables not represented here, or may not require the proposed SAN architecture at all.

These are not decorative disclaimers. They define the work required to turn the framework into a stronger theory.

17. Conclusion

Biological computationalism is most useful as a positive constraint. If cognition is not substrate-free code, a theory must identify what the substrate does. The answer cannot stop at the claim that biology is complex, continuous, energetic, or embodied. It must specify states, transitions, delays, noise, persistence, recurrence, measurement, and failure.

SAN contributes a candidate joined architecture. Ongoing biological activity supplies a context; arriving events are transformed relative to that context; state-dependent outputs alter downstream receivers; recurrent bodily and network consequences update the system again. The oscillatory canvas, PWD, and NAPOT name progressively more specific parts of that proposal only after the underlying problem and operation are clear.

The resulting claim is strong enough to test and narrow enough to defend. Biological dynamics may matter to consciousness because they perform causal work that an abstract terminal description omits. An artificial system is not excluded, but its equivalence must be earned through mechanism, dynamics, and intervention rather than assumed from surface behavior.

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